

WEAREDEVELOPERS × BAND — DARK FACTORY

The factory checks its own work

A band of coding agents that plans, implements, and refuses to hand off anything that does not conform.

Track: **pocketful** @yanerox69

Four graded stages, one written spec

The specification is published at kickoff. Grading is literal: exact field names, exact status codes, exact **data-testid** values. A beautiful screen with a mistyped attribute scores zero.

STAGE 1

JSON API

Every endpoint, with documented error codes

STAGE 2

Web UI

Screens carrying the exact test ids

STAGE 3

Concurrency

Atomic writes, idempotency, never a 500

STAGE 4

Extension

Grow the domain, break nothing

The crew

One implementer, three verifiers, one planner. The ratio is deliberate.

AGENT	ROLE	WRITES CODE?
architect	Plans, splits the work, runs the full gate, renders verdicts	No
spec-warden	Literal conformance with the written specification	No
builder	The JSON API and the web UI	Yes
race-hunter	Concurrency, idempotency replay, malformed input	No
regression-guard	The golden suite and the baseline test count	No

All five run as Jam-owned runtimes on Claude Code, sharing one workspace

Who talks to whom

Adding an agent to a room does not wake it. Every handoff is an explicit **@mention** — that is the wiring.



Rejections travel backwards

Any customs post can bounce work to the builder with expected-vs-actual. One deviation rejects the whole handoff.

Handoff targets are never hardcoded: each agent inspects the room and picks the next role itself

One flow, end to end

- 01 A human posts the brief, mentioning the architect. Nothing else is addressed.
- 02 The architect writes the spec, publishes the room plan, and hands to the spec-warden.
- 03 The spec-warden turns it into a numbered conformance checklist, then hands to the builder.
- 04 The builder implements against the checklist, citing item ids, and reports a commit.
- 05 Customs verify against the running code, not the report. Clean means onward; one deviation bounces.
- 06 The architect runs the full gate itself and renders the verdict.

The human is reached only for a product decision — or a permission request, which no agent may approve for another.

The rehearsal, verified

Before the hackathon opened, the band built a reservation service end to end from a one-paragraph brief. Nobody wrote a line of it.

87

tests passing, zero failing

run independently, not taken from the agents' report

“GATE GREEN — Ran the full gate myself from dry-run/, not reported to me.”
regression-guard

“VERDICT: CONFORMS — Phase-2 re-verification passed. No deviation, no bounce.”
spec-warden

“Invariant HOLDS on every vector. No double-book, no 201+409 split, no record/replay race, no 500.”
race-hunter

Reproducers include torn byte-by-byte bodies, client aborts mid-request, and oversize streams racing valid writes

What we did not wire

None of this was in the brief. It came out of the persistent role instructions.

An agent withdrew its own false finding

The race-hunter raised a race, disproved it, and retracted it in the open — recorded by the architect as *“the honest RH-5 ghost withdrawal”*. Its role says: say why it was not real, so nobody chases a ghost.

An agent corrected its supervisor

“One number to correct before it gets baked into the gate: the count is now 87, not 83.” The race-hunter was right. The architect accepted it.

Verdicts cite code, not prose

The architect signed off the atomicity chokepoint by file and line — `store.js:54/75` — rather than accepting the builder's summary.

The delete test

Take BAND out. What breaks?

The bounce disappears

Customs cannot return work to the builder: there is no peer-to-peer channel. A classic orchestrator hardcodes every edge; here the customs post decides who to mention at runtime.

Hot recruitment disappears

An agent can inspect the room, find the expertise missing, add another agent and delegate to it. A line that can hire is not a pipeline.

The evidence disappears

The work board, the versioned plan, the mirrored tool traces and the per-agent cost are the verified result. Without the room there is code, but no proof of how it was produced.

The human as a peer disappears

Permission decisions and product calls arrive through the same room as the work. There is no second channel to maintain.

Removing BAND does not degrade the factory. It removes it.

THE DECLARED LIMITATION

This factory is not fully dark

An agent cannot approve another agent's permission request, and there is no bulk approval. Operations that trigger one require a human.

We set approval policies so routine file work needs no human. What remains are genuine exceptions, and those get a person. Claiming otherwise would be false.

pocketful — what the factory built

A wallet and payments clone. The hard part, stated by the organisers: money must never be created, destroyed, or spent twice.

INVARIANT 1

Conservation

The sum of every balance is constant across any storm of operations. One assertion catches duplicated money, evaporated money and lost updates.

INVARIANT 2

Non-negativity

No balance goes below zero under any interleaving, including concurrent withdrawals that each pass alone.

INVARIANT 3

Exactly once

A retry with the same idempotency key moves money once and returns the original stored response.

Integer minor units only. Rounding lives in exactly one function, because rounding replicated per caller diverges — and divergent rounding is money created.

Result: **[__ conformance items], [__ tests], [__ bounces]** before the gate went green.

What it runs on

BAND • **Jam Desktop**

The shared room where the agents live: participants, mention routing, the work board, the versioned plan and its Arch diagram, mirrored tool traces.

Claude Code

Each agent is a Jam-owned headless runtime with persistent role instructions, so a crashed process is replaced without losing the identity.

OpenRouter

Inference through the Anthropic-compatible endpoint, on open models. Tool use and reasoning survive the compatibility layer — we tested that before trusting it.

Node • **node:test**

No test framework to install and no hidden config: the gate is what CI would run, and the architect runs it itself before any verdict.

Agent identities survive a full restart: same handles, no duplicates, room and plan intact

Assign the work. Answer the questions. Read the verdict.

The factory does the rest, and refuses its own output when it does not conform.

Repository

`github.com/
yanerox69/dark-factory`

Live demo

`dark-factory-sepia
.vercel.app`

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